Configuring Basic Network HardwareTheory

Practical Assignment

Aim

To learn how to access a Wi-Fi router, identify network settings, test a static IP address, and understand the router's DHCP range.

1. Open the Router Admin Login Page

Connect your laptop or mobile phone to your home Wi-Fi. Open a web browser and enter your router's address, commonly **192.168.0.1** or **192.168.1.1**. You can find the correct address by checking the **Default Gateway** in your device's network settings. The router login page should appear. Take a screenshot of the login page for your practical record. Do not share your router password.

Screenshot: Attach your own router login-page screenshot here.

2. Find Your Device's Network Settings

While connected to Wi-Fi, find and record these three values from your laptop or mobile device:

Setting	What it means	Your value
IP address	The address assigned to your device on the network.	
Subnet mask	Shows which part of the IP address identifies the local network.	
Default gateway	Usually the router's local IP address; it connects your device to other networks.	

On Windows: open Command Prompt and type **ipconfig**. Look for IPv4 Address, Subnet Mask, and Default Gateway under your Wi-Fi adapter. On a phone, open Wi-Fi settings and tap the connected network to view its details.

3. Test a Static IP Address

A static IP is entered manually instead of being assigned automatically by DHCP. First note your current settings. Choose an unused IP address in the same subnet as your router. For example, if the router is **192.168.1.1** and the subnet mask is **255.255.255.0**, an example address could be **192.168.1.50**—but only if it is not already in use and does not conflict with the DHCP pool. Use the same subnet mask, set the default gateway to the router's IP, and keep DNS settings as provided by your network or router.

On Windows, open Settings → Network & Internet → Wi-Fi → your connected network's properties → IP assignment → Edit. Select Manual, enable IPv4, enter the values, and save. Menu names can differ by Windows version. Test whether websites load. After the test, change IP assignment back to **Automatic (DHCP)** and save.

Important: If internet access stops, the IP may be incorrect, already in use, or outside the suitable range. Restore Automatic (DHCP). Do not change router settings unless you are authorised to manage the home network.

4. Find the Router's DHCP Range

Sign in to the router admin panel and look for a menu such as **LAN**, **Local Network**, or **DHCP Server**. Write down the DHCP starting and ending addresses shown by your router. For example, a range might

be 192.168.1.100 to 192.168.1.200. This is only an example; record the actual range from your router.

DHCP setting	Value from your router
DHCP start address	
DHCP end address	

What happens if two devices have the same IP address?

If two devices use the same IP address, an IP conflict can occur. One or both devices may lose network access, and connections can become unreliable. A DHCP server normally tries to avoid assigning the same address twice, but conflicts can happen if a manually configured static IP overlaps with the DHCP range.

Conclusion

This practical explains how to open a router's login page, check IP address information, test a static IP safely, and locate the DHCP range. After testing a static IP, the device should be returned to Automatic (DHCP).